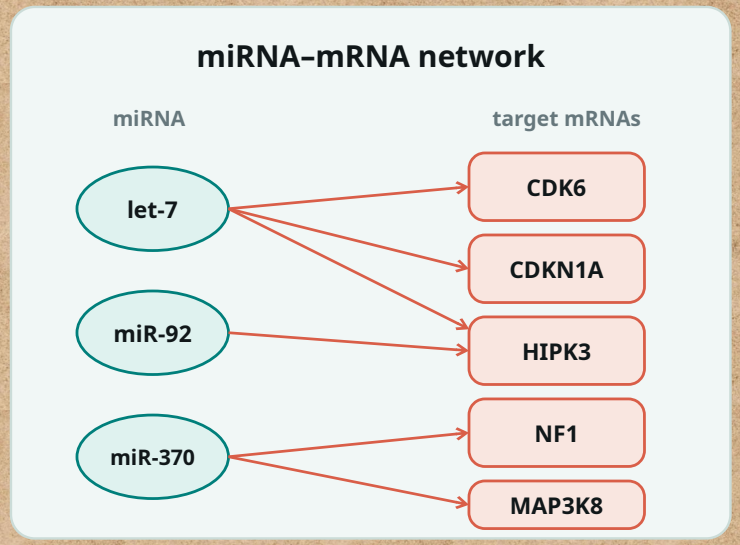
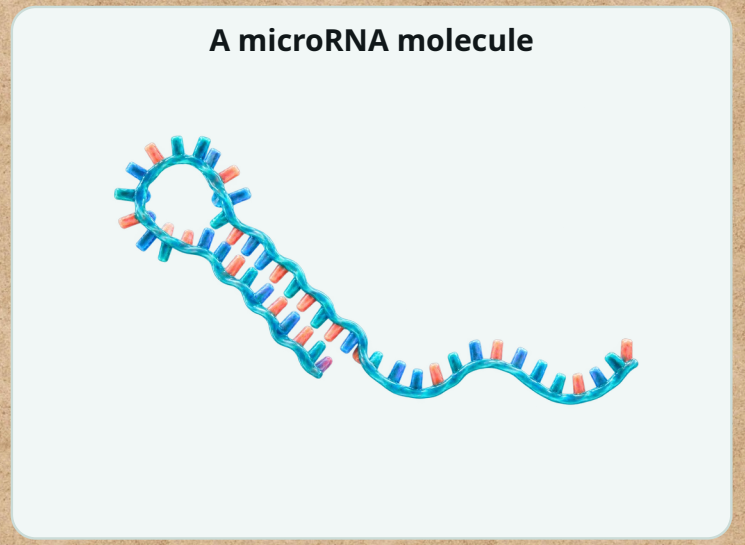
The sound of silencing

Background: microRNA (miRNA) silences gene expression by binding to messenger RNA (mRNA)

Questions: 1.) Do microRNAs regulate genes differently in humans and chimps as the brain develops?
2.) Can that help explain what makes the human brain special?







METHODS

- Time series analysis of weighted regulatory networks
- Functional analysis

DATA

Publicly available miRNA and mRNA expression data from a study

FIELDS

- Computational neuroscience and genomics
- General network analysis

Source data

A) SOURCE DATA AND DATA CLEANING

SOURCE DATA

Prefrontal cortex (PFC) expression data from

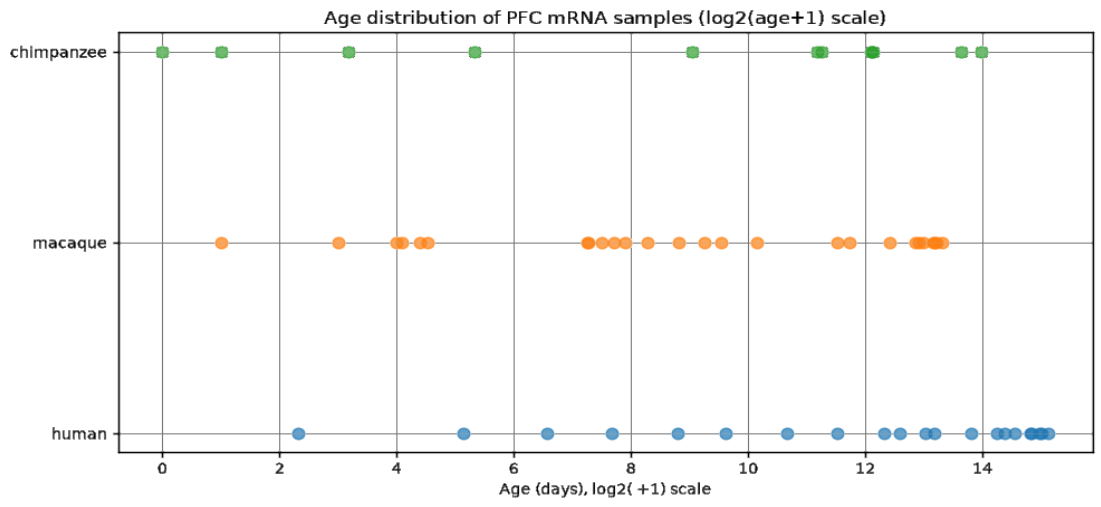
Somel et al. MicroRNA-Driven Developmental Remodeling in the Brain Distinguishes Humans from Other Primates. PLOS Biology 9(12) (2011).

DATA FILTERING AND CLEANING

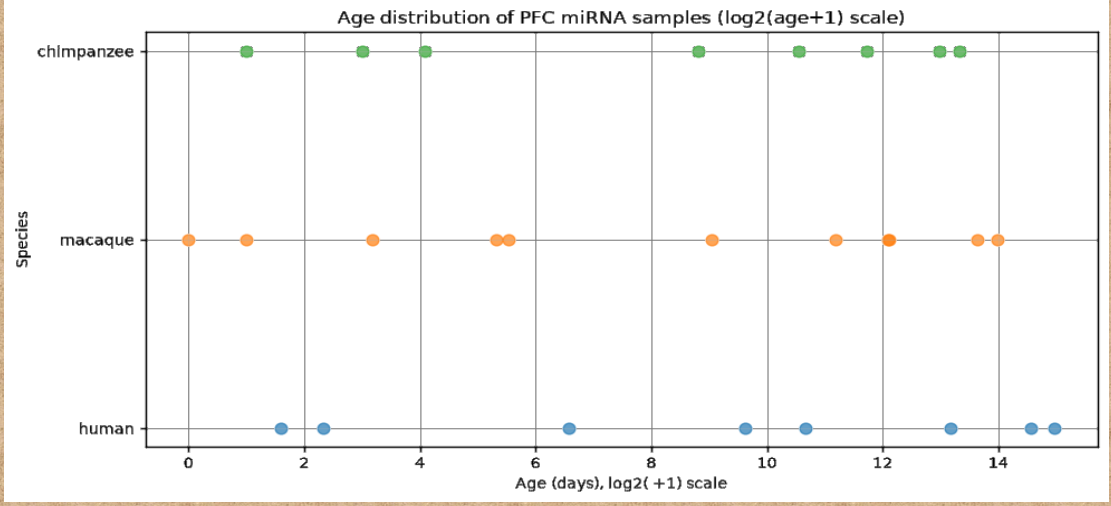
- Removed duplicate measurements, prenatal individuals and outliers
- Matched probes to genes
  Rule: retain 1 probe ↔️ 1 gene

RESULT

- mRNA: 58 retained profiles, 11,667 genes
- miRNA: 27 profiles, 198 detected miRNAs
- Human, chimpanzee, rhesus macaque



mRNA · $\log_2(\text{age} + 1)$



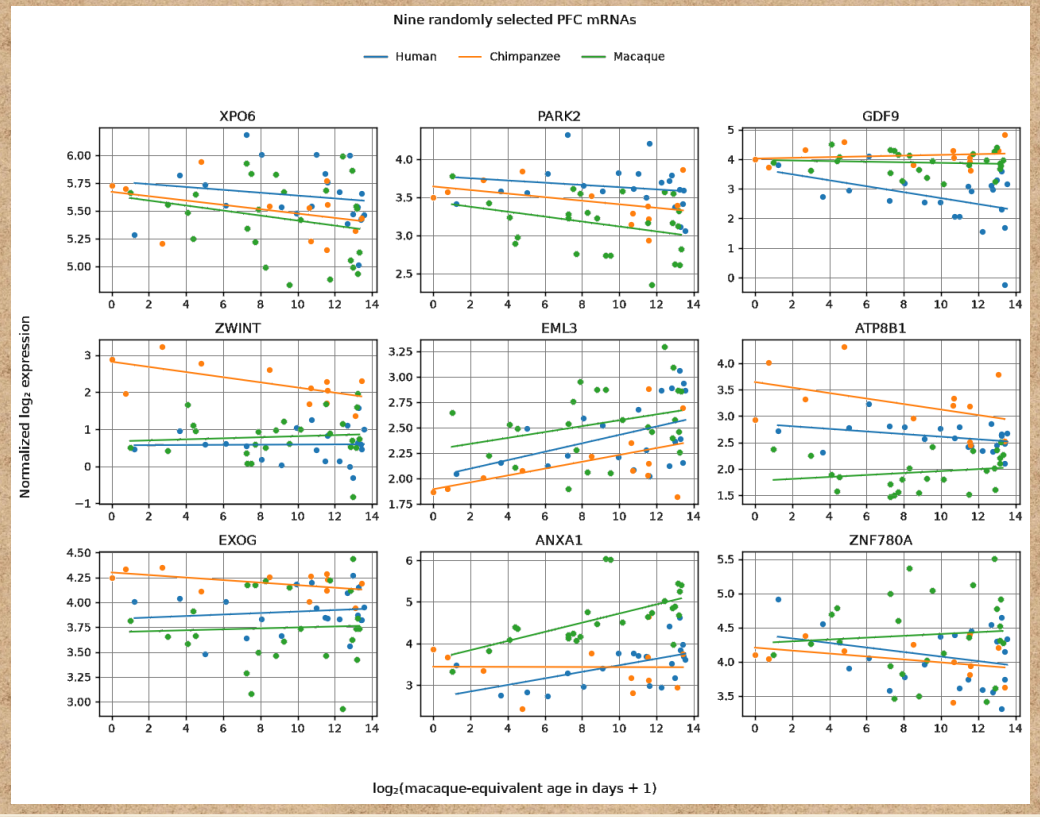
miRNA · $\log_2(\text{age} + 1)$

Exploratory data analysis

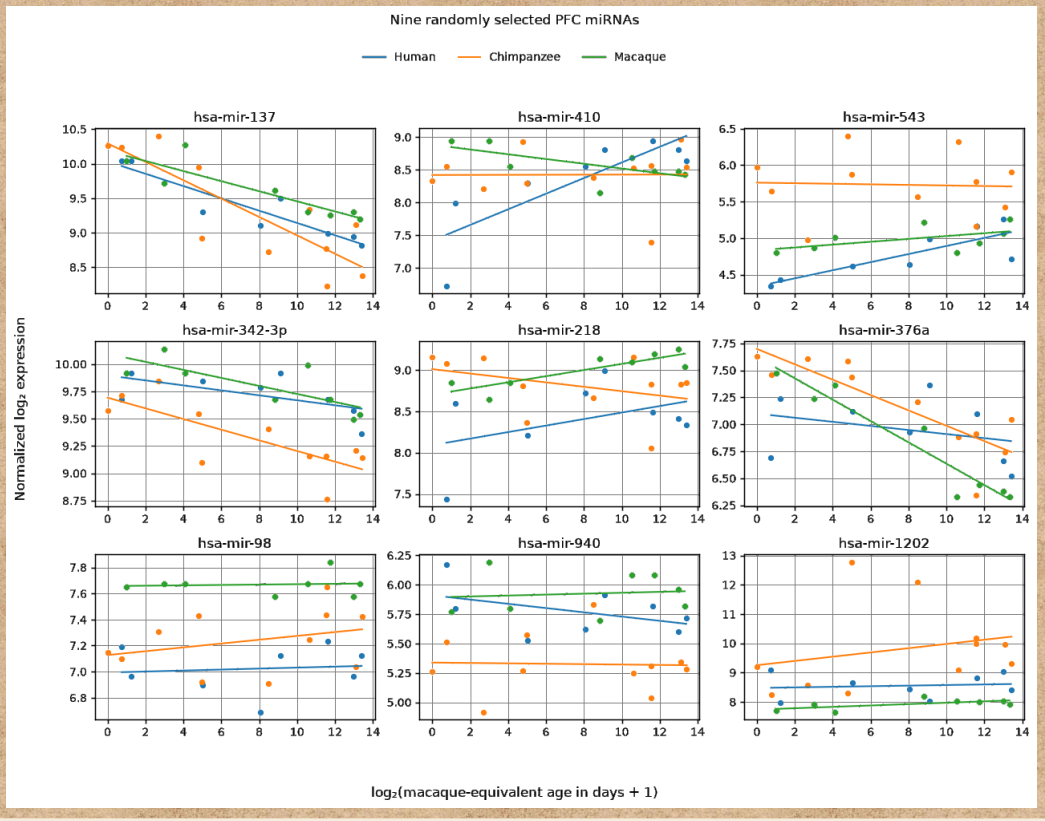
A) REGRESSION ANALYSIS

MODEL $\log_2(\text{expression}) \sim \text{species} + \log_2(\text{age} + 1)$

mRNA · nine example genes



miRNA · nine example miRNAs



RESULT: MEAN IN-SAMPLE R^2

mRNA: species 25.8% | age 13.0% | both 38.2%
miRNA: species 24.8% | age 23.9% | both 49.7%

Human–chimpanzee differences in early development

B) DIFFERENTIAL EXPRESSION

EARLY DEVELOPMENT

Nine profiles under 100 days:
5 human and 4 chimpanzee.

TOOL

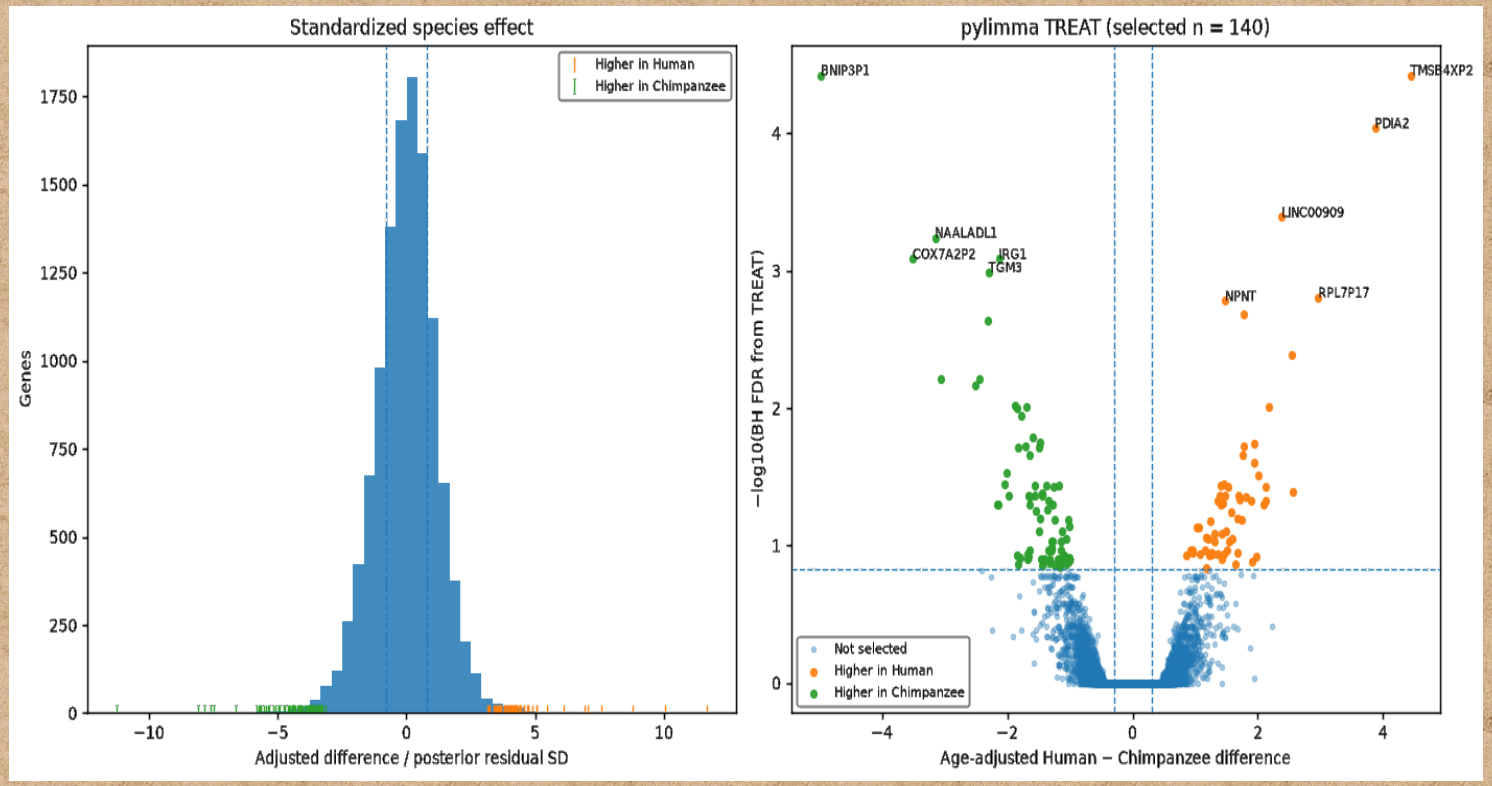
pylimma: Linear Models for Microarray Data.

RESULT

140 of 11,667 genes selected:
62 higher in humans; 78 in chimps.

SIGNIFICANCE & ROBUSTNESS

Age-adjusted TREAT FDR < 0.15,
$|\log_2$ difference$|$ ≥ 0.30, plus effect-size and leave-one-out direction checks.



Standardized species effect and TREAT FDR

Autism genes in early development

B) SFARI AUTISM-GENE ENRICHMENT

QUESTION

Are autism genes enriched among the 140 early-expression candidates?

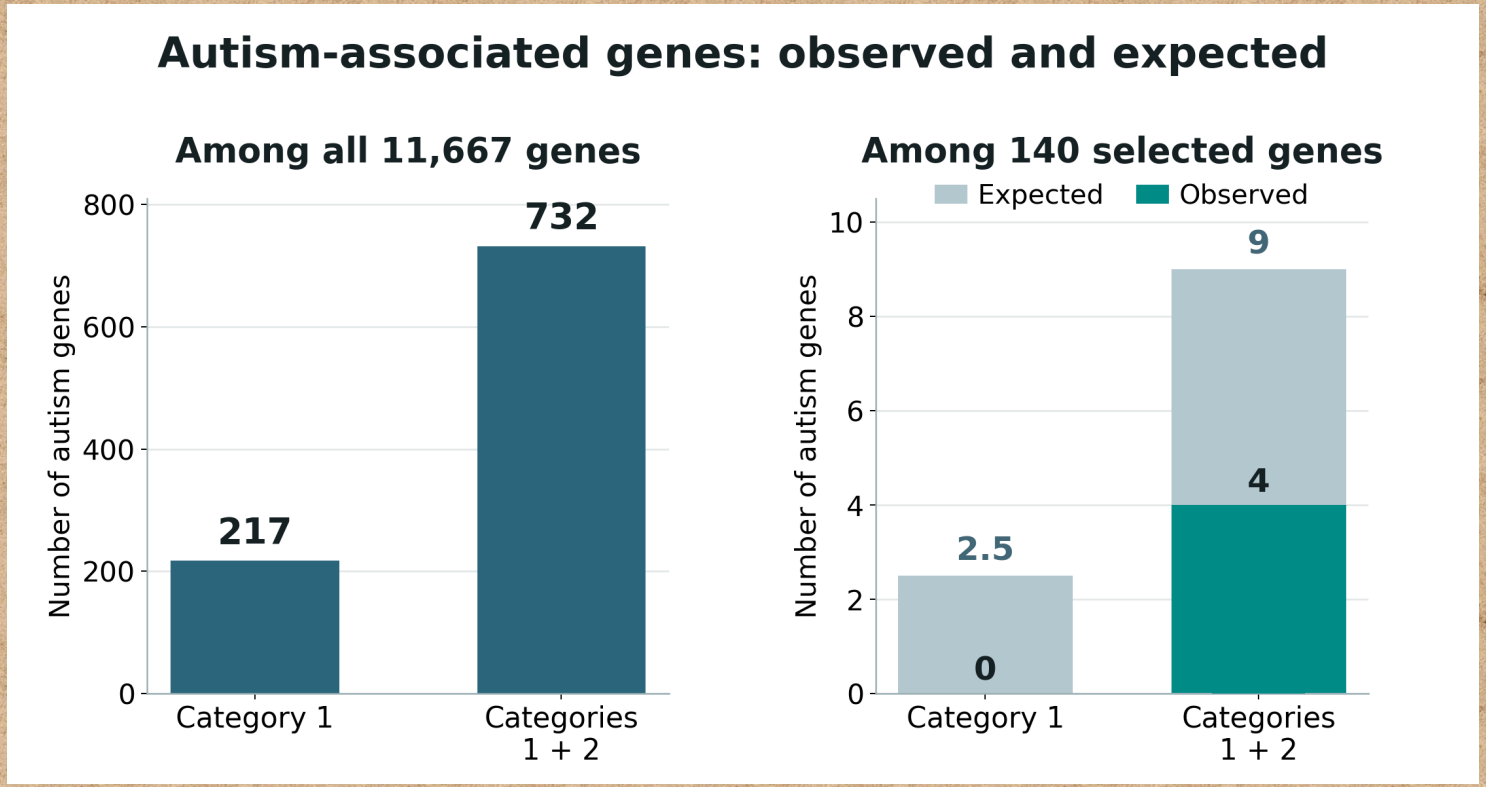
FISHER EXACT TEST (ONE-SIDED)

Category 1: 0 / 140
All tested genes: 217 / 11,667
p = 1.000

Categories 1 + 2: 4 / 140
All tested genes: 732 / 11,667
p = 0.979

CONCLUSION

Neither SFARI set was enriched.
The tests do not prove depletion.



Gene Ontology (GO): functional analysis

C) GO BIOLOGICAL PROCESS ANALYSIS

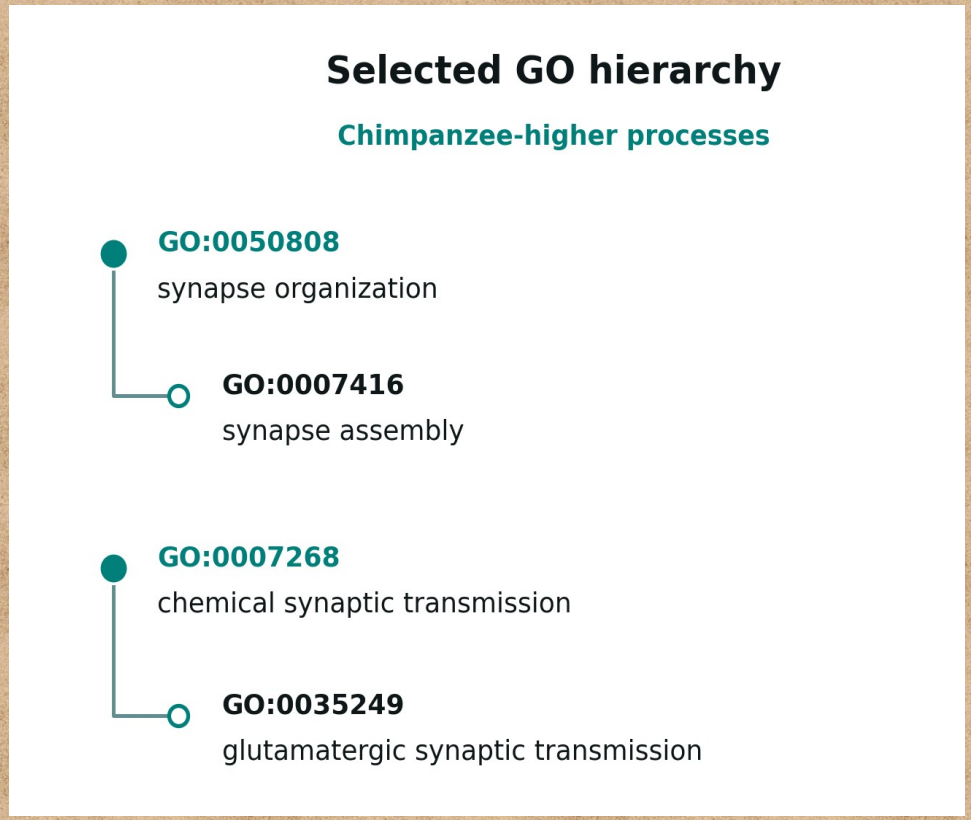
METHOD

Ranked differential-expression statistics with GOfuncR;
10,386 annotated genes,
13,796 Biological Process terms tested.
FWER ≤ 0.05 (10,000 random sets).

RESULTS AFTER HIERARCHY REFINEMENT

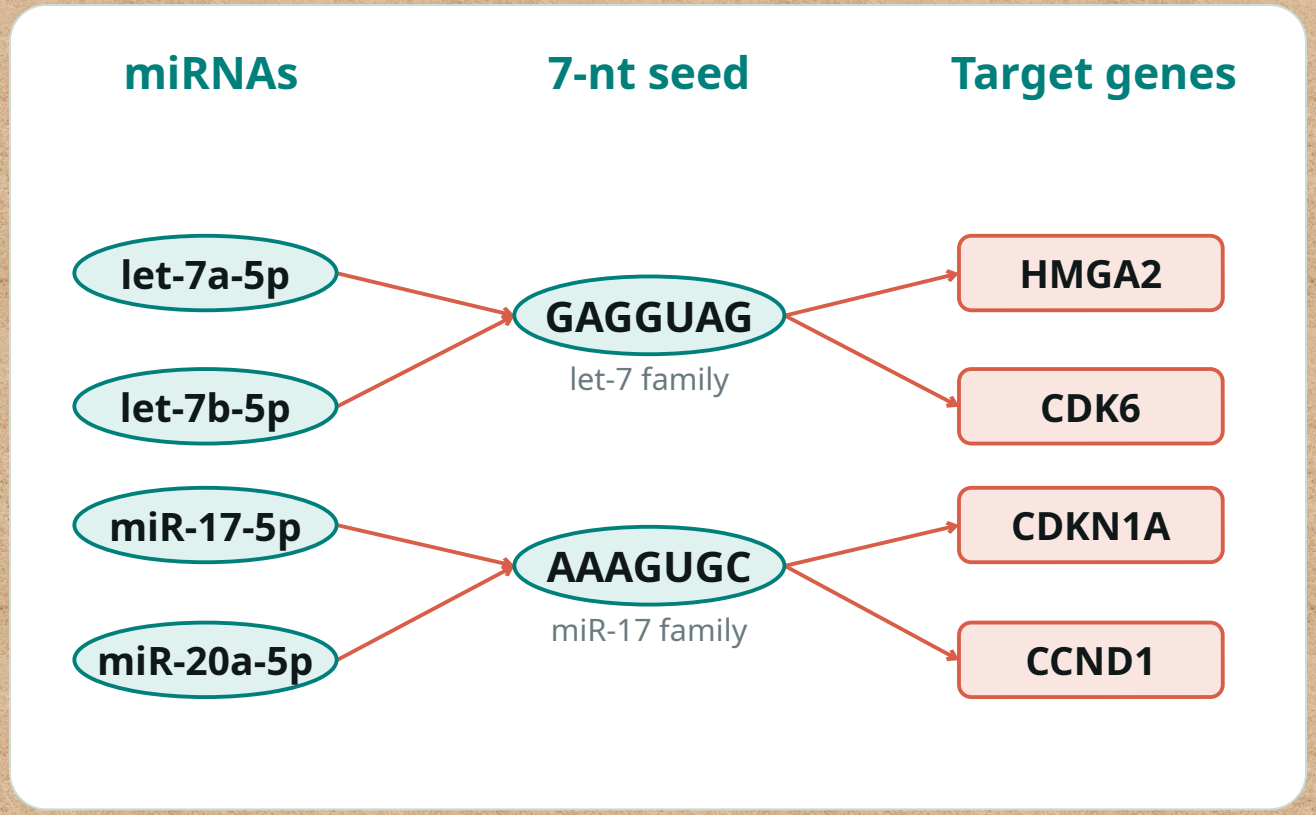
10 human-higher GO categories: transcription, RNA splicing, DNA processes.

30 chimp-higher GO categories: synaptic function, ion transport and mitochondrial electron transport.



Life-history network analysis

D) LIFE-HISTORY NETWORK ANALYSIS



CONSTRUCTION

Measured miRNAs → seven-nucleotide seed families → miRDB predicted targets

NETWORK SIZE

24,111 seed–gene edges; 156 seeds and 7,039 target genes.

AGE-MATCHED PROFILES

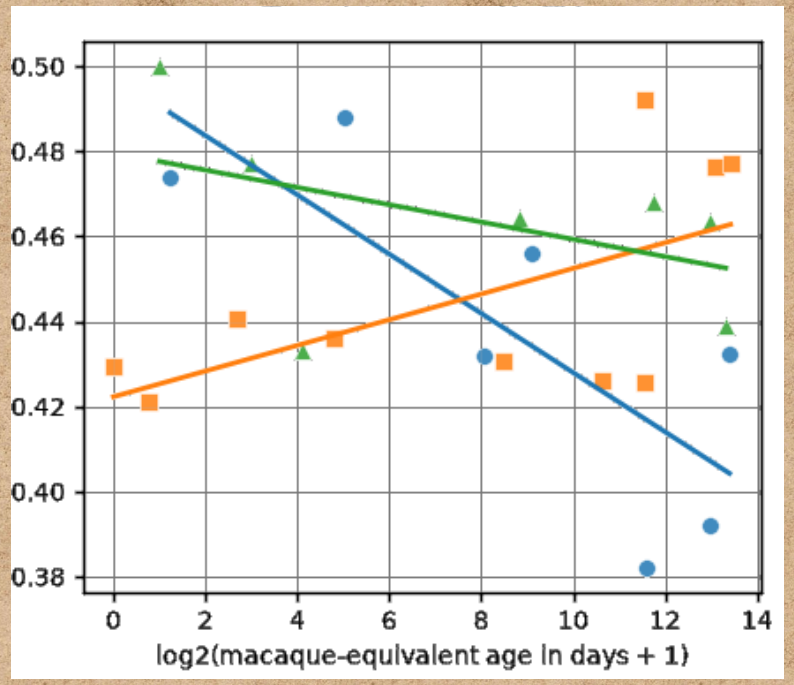
24 profiles: 7 human, 10 chimpanzee, 7 macaque. Edge scores combine standardized seed and target expression.

Life-history network results

D) SPECIES × DEVELOPMENTAL AGE

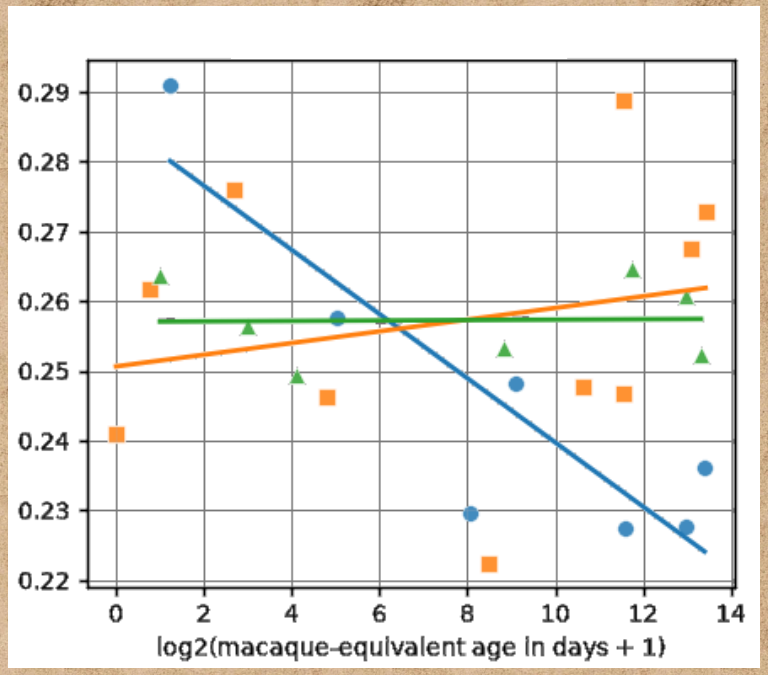
METHOD 17 matched human–chimpanzee profiles; life-history-aligned age; 5,000 permutations and BH-FDR across six features.

TARGET IN-STRENGTH CV



Δ slope = −0.01003 | FDR = 0.0132

EDGE-SCORE DISPERSION



Δ slope = −0.00544 | FDR = 0.0456

MAIN RESULT

Both measures decline more steeply with age in humans than in chimps.
Two of six network features were significant (FDR < 0.05).

Summary and contribution

E) MAIN FINDINGS AND DISTINCTIVE APPROACH

1.) EARLY HUMAN–CHIMP DIFFERENCES

140 genes (62 human-higher; 78 chimp-higher).

AUTISM-GENES

No enrichment of SFARI Category 1 or 1 + 2 genes

GENE ONTOLOGY

10 human-higher categories:
transcription, RNA splicing, DNA processes.

30 chimp-higher categories:
synaptic function, ion transport and mitochondrial electron transport.



2.) NETWORK LIFE-HISTORY ANALYSIS

With age, regulatory activity became more evenly distributed across target genes in the human samples.

SCOPE OF NOVELTY

Instead of analyzing network changes, we kept the network fixed and analyzed global network parameters using regression.